

Continuous Distillation

Final Lab Report
Unit Operations Lab 3
03/02/2020

Group 2, Tuesday, PM section

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Executive Summary:

The purpose of this experiment is to study the behavior of separation of binary mixture of water and ethanol using binary distillation column operating continuously at steady state. In addition, the feed position of the ethanol mixture was inserted in the midpoint, between tray 4 and tray 5 for the first half of the experiment, then moved to the top of the distillation column on tray 1 to compare found results. The found differences between the feeds was that the efficiency of the column, where the efficiency 36.4% for the midpoint feed and the top feed being 31.3%, showing that the appropriate column to use is the midpoint feed to obtain a efficient column.

An important process variable in this experiment is the reflux ratio at the top of the column. Furthermore, it is essential role in controlling the product purity and in determining the energy which is required in the reboiler. Understanding the effects of column plate efficiency and the number of stages is the goal for this experiment. Thus, the collected data will be graphed on McCabe-Thiele curve for Binary Distillation to explore the effects of the changing process variables being the position of the inlet feed with constants: reflux, pump power, trays (8), time. The Fenske is used to determine the minimum number of theoretical trays to find the efficiency of the column. The results were that the mid-inlet feed efficiency is 36% while the top-feed setup was 31% efficient.

In addition, understanding how continuous distillation columns operate gives incite on how chemicals like ethanol, common gas, diesel, and jet fuel are used to separate mixed feed streams into their own distinct products through a continuous distillation. The differences in set

up are important when understanding how a process could be improved, for example if a

Unit Operations ChE-381 Group No. 2

Tuesday, PM section

Continuous Distillation

University of Illinois

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column's efficiency is needed to be improved it would be a good idea to check the inlet feed to see if it is placed too high and possibly needed to be lowered closer to the midsection.

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Introduction:

In general, distillation is defined as the process in which a mixture is separated into its pure components [1]. This process plays an important role in chemical engineering, especially in petrochemical and refinery industries. Therefore, in chemical process plants, a distillation column is considered as one of the main unit operation [1]. Atmospheric distillation of crude oil is an example of distillation process in petrochemical industry. Crude oil is a petroleum product consist of a mixture of volatile liquid hydrocarbons [2]. During the atmospheric distillation process, the unrefined crude oil is separated into different fractions based on range of boiling point for each component which include gases, Naphtha, Kerosene, gas oil, and atmospheric residue [3]. Distillation process is also commonly used in liquor industry in order to make fruit spirits [4]. Due to the fact that, spirits are mostly consist of ethanol and water, distillation of binary mixture is one of the most common process in liquor industry [4]. Furthermore, distillation process can be categorized based on four main types which are simple distillation, steam distillation, fractional distillation, and vacuum distillation [5]. For the purpose of this experiment, a continuous-fractional distillation shown in **figure 1** was used in order to separate a mixture of ethanol and water.

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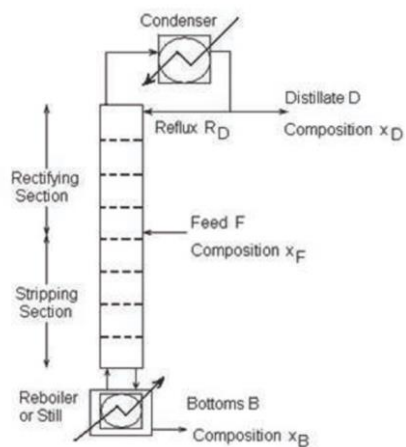


Figure 1: general Schematic of a distillation column [6].

In fact, fractional distillation is used to separate a mixture of ethanol-water due to the fact that these two components have different boiling points. Ethanol (C_2H_5OH) has a boiling point at $\sim 78.5^\circ C$ which is considered lower than water (H_2O) that has a boiling point $\sim 100^\circ C$. Thus, a mixture of these two components will boil at boiling point within the range of ($78.5^\circ C - 100^\circ C$). In general, during the process of separating ethanol-water mixture, most of the ethanol vaporizes in the top of the column, and it condenses as it cools down as shown in **figure 1** while water remains liquid in the bottom of the column.

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Commented [AZ11]: Briefly, what will you be doing in this experiment?

Theory:

The theory behind the continuous distillation column process is when the feed goes to the column through the middle stages (between the 4th and the 5th trays), the heat is being generated in the reboiler to heat up the mixture that contains ethanol and water. The boiling allows the product to vaporize to the top of the column, where it gets condensed and change phase to be liquid. This change of phases is what leads to the separation process to happen, where the vapor phase will be at the top and liquid phase will be at the bottom of the column. To control the product purity, the reflux ratio would be the variable to control and that is done by controlling the energy delivered to the reboiler.

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McCabe-Thiele Analysis for Binary Distillation

The graphical technique of McCabe-Thiele may be used to design the column and to test the effect of the process variable. This technique comes with many assumptions, and in order to use it in the right way, many conditions have to be applied. First, a diagram of the VLE (Vapor-Liquid-Equilibrium) has to be plotted and carefully looked at. Second, the column operating line that represent the construction of a stage to stage mass balance, which needs the liquid and the gas overall flows, also the reflux ratio. E.g. of McCabe-Thiele diagram:

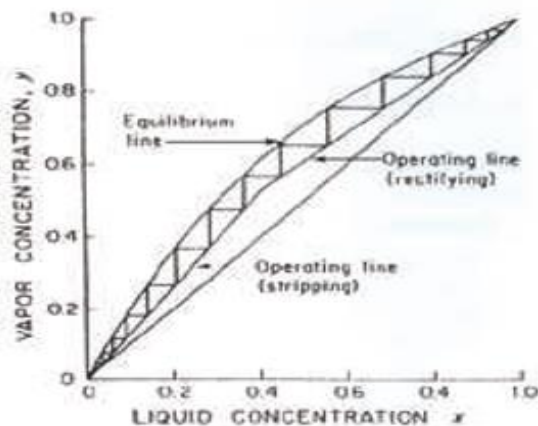


Figure 2. McCabe-Thiele diagram.

A continuous distillation column would have different operating lines (mass balances), one for the top part of the column and the other is for the bottom part of the column, where the feed comes in between these two parts. On the other hand, only one operating line can be found in a batch distillation and that is due to the constancy of the molar flows for liquid and vapor in the whole column. Number of stages can be determined by drawing a line that touches the equilibrium line to the operating line and by the end of every angle one can tell it is a stage.

Column Efficiency

The efficiency of the column can be found using this equation:

$$\eta = \frac{y_n - y_{n+1}}{y_n^* - y_{n+1}} \quad \text{Eqn. 1}$$

Where:

y_n , is the composition of vapor

X_n , is the composition of liquid

y_n^* , is the composition in equilibrium with X_n

If the assumption of the stages being operating under Vapor-Liquid-Equilibrium was true, then y_n would be equal to y_n^* . However, equilibrium is never attained entirely in practical and in many circumstances the assumption of achieving equilibrium in the stages is not applicable.

Results:

The found results were that the top plate setup will give higher results of EtOH composition this shows that if a higher needed volatile composition is needed then the higher the feed is placed in a distillation column. The two setups can be compared in figure 3 below, while the composition can be viewed the efficiency cannot be determined by the graph.

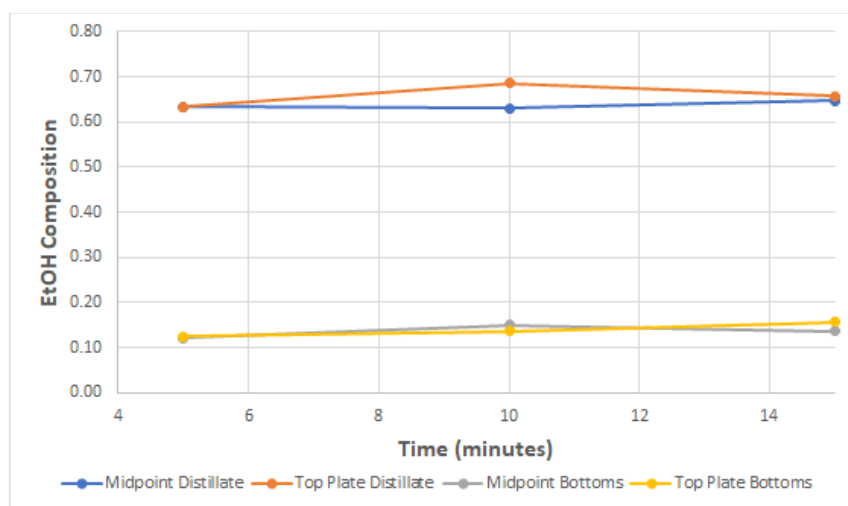


Figure 3: Time vs EtOH composition of top and midpoint setup

Furthermore, the composition of the wanted volatile product stream of the distillation column is used to determine the efficiency of the unit. To determine the efficiency of both setups the basic McCabe-Thiele method assumes vapor-liquid equilibrium on each stage [6] but for this experiment the Fenske equation was used to determine the theoretical plates. The found theoretical

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Commented [T19]: A higher NEEDED volatile composition if NEEDED? This doesn't make any sense

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plates were found to compare with the actual number of plates. The found results are shown below on **table 1**.

Table 1: Volatility and column efficiency of top and midpoint setup

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Distillation Column Estimated Design-Midpoint Feed			
Relative Volatility (Average):	1.853	Column Efficiency:	36.362
Theoretical Plates:	2.909	Actual Number of Trays:	8.000
Distillation Column Estimated Design-Top of the Column Feed			
Relative Volatility (Average):	2.028401	Column Efficiency:	31.33014
Theoretical Plates:	2.506411	Actual Number of Trays:	8

Discussion:

The purpose of this experiment was to examine a continuous distillation for separation 10 wt% solution of ethanol in water with a continuous inlet flow of 30% EtOH. Performing the steady state in this experiment by plotting McCabe- Thiele analysis for an ideal method. As for real data it was collected in the experiment a comparison made as shown in the result between real and ideal. The efficiency of each stage was analyzed for the tray composition, also for the stages regarding the column. This will lead to determine the overall efficiency for the column. Observing the found results of 31% for the top feed and 36% for the mid-feed. Having a higher efficient column is important for example when thinking of the operation cost of a unit it is important to have a higher efficiency to minimize the cost of running the operation.

In the continuous batch the vapor was raising up in the column then goes into the condensing process. Once verifying the steady state operation for the continuous batch distillation an assumption been made the vapor-liquid equilibrium determine the theoretical number of trays.

In the continuous batch distillation, when a liquid mixture exposed to temperature and reaches the boiling point, the composition for the vapor and the composition for the liquid are different. If the vapor been exposed with a separation and condensed to become a liquid, it will be richer for the boiling point of the initial mixture.

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This discussion section lacks major details: does your experiment agree with theory? What happened when you do the McCabe-Thiele analysis? How did your efficiency change when you change the feed position and why does it change the way it did? Sources of error?

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Commented [T26]: does this agree with theory? Why? Why not?

Commented [T27]: What does this mean to your experiment?

References:

- [1] Truong, H.S. & Ismail, Idris & Razali, Radzuan. (2010). Fundamental modeling and simulation of a binary continuous distillation column. 2010 International Conference on Intelligent and Advanced Systems, ICIAS 2010. 1 - 5. 10.1109/ICIAS.2010.5716218.
- [2] Chen, J. (2019). *Crude Oil*. [online] Investopedia. Available at: <https://www.investopedia.com/terms/c/crude-oil.asp> [Accessed 2 Mar. 2020].
- [3] Fahim, Mohamed A et al. *Fundamentals Of Petroleum Refining*. Elsevier, 2010.
- [4] Spaho, N. (2017). *Distillation Techniques in the Fruit Spirits Production*. [online] Available at: https://www.researchgate.net/publication/318073789_Distillation_Techniques_in_the_Fruit_Spirits_Production [Accessed 2 Mar. 2020].
- [5] Helmenstine, A. (2020). *Understand How Distillation Works*. [online] ThoughtCo. Available at: <https://www.thoughtco.com/what-is-distillation-601964> [Accessed 2 Mar. 2020].
- [6] Zdunek, Alan. *Continuous Distillation*, Student Lab Manual, 2020

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Appendix I: Data Tabulation/Graphs

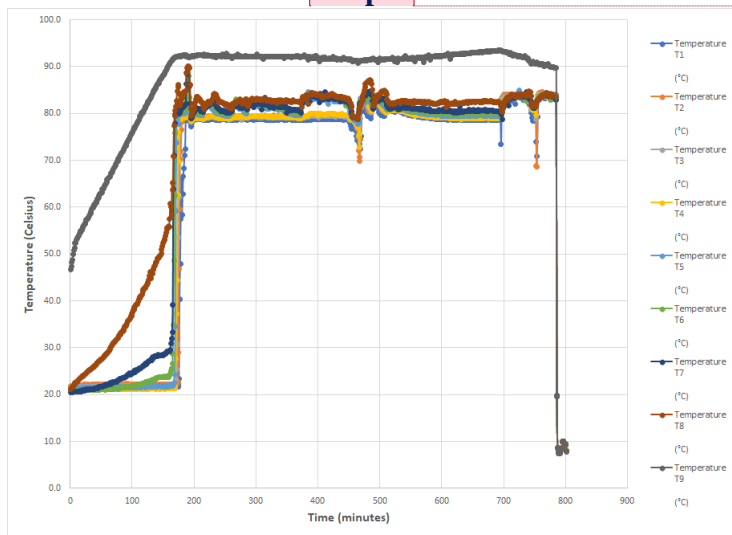


Figure 4: Time vs Temperature

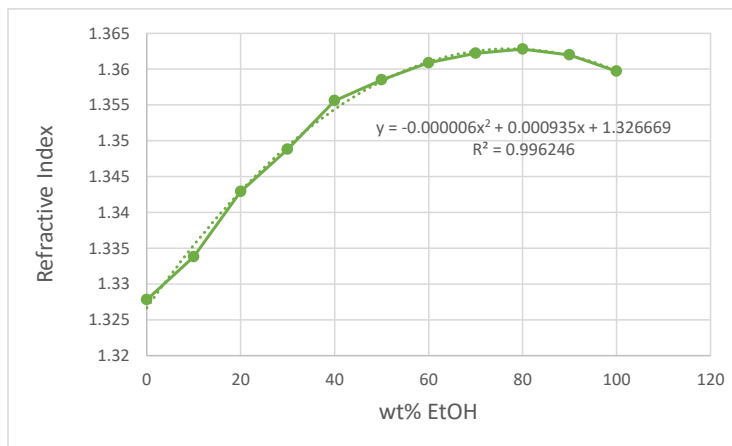


Figure 5: McCabe-Thiele Curve for Ethanol and Water

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Table 2: Feed Data for Both Setup

Above Top Plate Feed Position			
Distillate Summary: Composition of Ethanol		Bottom Summary: Composition of Ethanol	
Trials Average	0.6588	Trials Average	0.1392
Xdist EtOH	0.4302	Xdist EtOH	0.0595
Xdist H ₂ O	0.5698	Xdist H ₂ O	0.9405
Ydist EtOH	0.6252	Ydist EtOH	0.3371
Ydist H ₂ O	0.3748	Ydist H ₂ O	0.6629
Relative Volatility (a)	0.4526	Relative Volatility (a)	9.0915
Midpoint Feed Position			
Distillate Summary: Composition of Ethanol		Bottom Summary: Composition of Ethanol	
Trials Average	0.6370	Trials Average	0.1360
Xdist EtOH	0.4069	Xbot EtOH	0.0580
Xdist H ₂ O	0.5931	Xbot H ₂ O	0.9420
Ydist EtOH	0.6162	Ybot EtOH	0.3309
Ydist H ₂ O	0.3838	Ybot H ₂ O	0.6691
Relative Volatility (a)	0.4273	Relative Volatility (a)	8.0366

Distillation Column Estimated Design-Mipoint Feed			
Relative Volatility (Average):	1.853	Column Efficiency:	36.362
Theoretical Plates:	2.909	Actual Number of Trays:	8.000
Distillation Column Estimated Design-Top of the Column Feed			
Relative Volatility (Average):	2.028401	Column Efficiency:	31.33014
Theoretical Plates:	2.506411	Actual Number of Trays:	8

Top Feed			
Top Product			
Sample	1	2	3
Refractive Index	1.3622	1.3621	1.3625
Sample #	382	414	447
Temperature (F)	78.8	78.8	78.8
Boiler Temp	92.1	91.6	91.6
Bottom Product			
Sample	1	2	3
Refractive Index	1.3373	1.3394	1.3391
Sample #	383	413	446
Temperature (F)	83.7	83.8	83.1

Mid Feed			
Top Product			
Sample	1	2	3
Refractive Index	1.3622	1.3633	1.3627
Sample #	720	752	782
Temperature (F)	83.6	80.3	83.3
Boiler Temp	92.5	90.7	89.8
Bottom Product			
Sample	1	2	3
Refractive Index	1.3376	1.3393	1.3399
Sample #	720	752	782
Temperature (F)	83.6	81.7	83.7

Appendix II: Error Analysis

During the experiment there were multiple possibilities for error to occur. For instance, the calibration procedure was done for the feed pump in order to eliminate instrumental uncertainties and obtain the accuracy [6]. This type of error is considered a systematic error in which the effect on the obtained data can be significant if not done properly. A major error that could occur is mistaking the direction to open the bottom valve [V2] because one direction would allow to collect the sample and the other one is to drain the tank. Draining the tank would lead onto starting over the whole experiment which require so much time. Lastly, during the experiment due to human error this occurred.

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Commented [AZ32]: Expand this section to include quantitative analysis; you obtained 3 samples, so you can calculate the std dev in your measurement.

Appendix III: Sample Calculations

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Feed Pump Volumetric Flowrate

After plotting the volumetric flow rate against the pump speed setting, linear graph below was obtained:

$$Y = 26.636X - 4 \quad \text{Eqn. 2}$$

The pump was set to 2 for this experiment. Applying Eqn.1 we find the feed rate to the column.

$$Y = 26.636(2) - 4 = 49.273 \text{ mL/s}$$

Weight% Ethanol from Refractive Index

After plotting the refractive index against known weight% ethanol solutions the linear graph below was obtained:

$$Y = (-6 * 10^{-5})X^2 + (9.35 * 10^{-4})X + 1.326669 \quad \text{Eqn. 3}$$

Calculate the calculate the weight% of one of the column distillates samples with a refractive index of 1.3622 by applying Eqn 2.

$$Y = -6 * 10^{-5}(1.3622)^2 + (1.3622) * 9.35 * 10^{-4} * +1.326669$$

$$Y = 60.40\%$$

Relative Volatility

$$Rel. Volatility = \frac{X_{EtOH}/Y_{EtOH}}{X_{H2O}/Y_{H2O}} \quad \text{Eqn. 4}$$

The mole fraction vaport-Liquid equilibrium data for ethanol and water at 1atm was used for this calculation

The weight% composition of the distillate sample is taken and converted to moles.

$$mole \text{ of ethanol} = \frac{Weight\% \text{ ethanol}}{Molecular \text{ weight of ethanol}} \quad \text{Eqn.4}$$

$$= 0.6588 * \frac{Mol}{46.07 \text{ g}} = 0.01429$$

$$mole \text{ of water} = \frac{1 - Weight\% \text{ ethanol}}{Molecular \text{ weight of water}} \quad \text{Eqn.5}$$

$$= 1 - 0.6588 * \frac{Mol}{18.015 \text{ g}} = 0.08939$$

Then the mole fractions are calculated

$$X_{distEtOH} = \frac{\text{Moles of ethanol}}{\text{Moles of solution}} \quad \text{Eqn. 6}$$

$$X_{EtOH} = \frac{0.01429}{0.01429 + 0.08939} = 0.4302$$

From the VLE data interpolate to calculate the liquid and vapor mole fraction of water in the distillate. Use Eqn. 3 to calculate the relative volatility by plugging in the values for the vapor and liquid molar fractions.

Theoretical Number of Plates

To calculate the theoretical number of plates needed for the column the Fenske Equation was used

$$N = \frac{\log \left[\left(\frac{X_d}{1-X_d} \right) \left(\frac{1-X_b}{X_b} \right) \right]}{\log \alpha_{avg}} \quad \text{Eqn. 7}$$

where:

- N is the minimum number of theoretical plates required at total reflux (of which the reboiler is one)
- X_d is the mole fraction of more volatile component in the overhead distillate
- X_b is the mole fraction of more volatile component in the bottoms
- α_{avg} is the average relative volatility of the more volatile component to the less volatile component

Column efficiency

Eqn. 1 from the theory section of this report was used to calculate the efficiency of the column

$$\eta = \frac{y_n - y_{n+1}}{y_n^* - y_{n+1}} \quad \text{Eqn. 1}$$

Appendix IV: Individual Team Contributions

NAME: Ahmad Abusubaih

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8	
Pre-Lab Editing	3	
Experimental Objective (Prelab)	0	
Apparatus (prelab)	3	
Materials & Supplies (prelab)	1	
Procedure (prelab)	0	
Project Safety Evaluation (prelab)	1	
Final Lab Editing	3	
Executive Summary (final lab)	0	
Introduction (Final lab)	0	
Theory (Final Lab)	1	
Results (final lab)	1	
Discussion (final lab)	3	
Conclusion (final lab)	0	
References (final and/or prelab)	1	
Data Tabulation/Graphs (final)	2	
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	1	
Power Point Presentation	1	
Total	29	

NAME: Fatimah Al Ibrahim

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8	
Pre-Lab Editing	1	
Experimental Objective (Prelab)	1	
Apparatus (prelab)	1	
Materials & Supplies (prelab)	1	
Procedure (prelab)	1	
Project Safety Evaluation (prelab)	1	
Final Lab Editing	1	
Executive Summary (final lab)	1	
Introduction (Final lab)	1	
Theory (Final Lab)	1	
Results (final lab)		
Discussion (final lab)		
Conclusion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)	1	
Error Analysis (final lab)	1	
Sample Calculations (final Lab)	1	
Power Point Presentation	1	
Total	22	

NAME: Ahmed Alghamdi

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8	
Pre-Lab Editing	1	
Experimental Objective (Prelab)	1	
Apparatus (prelab)	1	
Materials & Supplies (prelab)	1	
Procedure (prelab)	1	
Project Safety Evaluation (prelab)	1	
Final Lab Editing	1	
Executive Summary (final lab)	1	
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)	1	
Discussion (final lab)	1	
Conclusion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)		
Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation	1	
Total	19	

NAME: Velia Palma

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8	
Pre-Lab Editing	3	
Experimental Objective (Prelab)	3	
Apparatus (prelab)		
Materials & Supplies (prelab)	1	
Procedure (prelab)	2	
Project Safety Evaluation (prelab)		
Final Lab Editing	1	
Executive Summary (final lab)	1	
Introduction (Final lab)		
Theory (Final Lab)	2	
Results (final lab)		
Discussion (final lab)		
Conclusion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)	3	
Error Analysis (final lab)		
Sample Calculations (final Lab)	3	
Power Point Presentation		
Total	27	

NAME: Christian Sanchez

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	5	
Pre-Lab Editing	2	
Experimental Objective (Prelab)	1	
Apparatus (prelab)	2	
Materials & Supplies (prelab)	2	
Procedure (prelab)	2	
Project Safety Evaluation (prelab)	2	
Final Lab Editing	2	
Executive Summary (final lab)	2	
Introduction (Final lab)	1	
Theory (Final Lab)	1	
Results (final lab)	3	
Discussion (final lab)	1	
Conclusion (final lab)	1	
References (final and/or prelab)	1	
Data Tabulation/Graphs (final)	1	
Error Analysis (final lab)	1	
Sample Calculations (final Lab)	1	
Power Point Presentation	0	
Total	29	

